

# Real-Time ML Inference Server

Maharshi Soni | [model-serving](#) | [advanced](#)

## Description

---

FastAPI + WebSocket real-time ML inference server with batched prediction, model hot-reloading, A/B testing, health monitoring, and in-memory feature cache.

## Tech Stack

---

fastapi, uvicorn, websockets, scikit-learn, numpy, pydantic, httpx, pytest

## Features

---

- WebSocket streaming predictions
- REST batch inference
- Model hot-reloading
- A/B testing
- Feature cache with TTL
- Health monitoring

## Sample Input

---

Feature vectors

```
curl -X POST /predict -d '{"features": [1.2, 3.4, 5.6]}'
```

## Sample Output

---

Predictions with metadata

```
{"prediction": 1, "probability": 0.87, "model": "v2"}
```

## Project Files

---

```
.gitignore
main.py
project-idea.json
pyproject.toml
README.md
requirements.txt
.github/workflows/test.yml
models/challenger_model.joblib
models/demo_model.joblib
src/ab_testing.py
```

```
src/batch_queue.py
src/cache.py
src/config.py
src/model_registry.py
src/monitoring.py
src/schemas.py
src/server.py
src/train.py
src/__init__.py
tests/conftest.py
tests/test_ab_testing.py
tests/test_cache.py
tests/test_model_registry.py
tests/test_monitoring.py
tests/test_server.py
tests/__init__.py
```

## Test Results

---

**PASSED**

Run: 31 | Passed: 31 | Failed: 0

*Maharshi Soni | MIT License*